Structural Entropy Dark Energy against DESI DR2, Pantheon+ and Planck 2018: a zero-extra-parameter test

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Abstract

We confront Structural Entropy Dark Energy (SEDE) — a structure-gated, H-linear horizon-entropy dark energy realized as a cubic kinetic-gravity-braiding theory with braiding fraction b = 0.20 — with the full DESI DR2 BAO, Pantheon+ SN, and Planck 2018 (lowl TT + high-ℓ TTTEEE-lite) likelihoods, computed through the mochi-class modified-gravity Boltzmann code inside cobaya. SEDE's background w(z) is fixed by its reconstruction, so at the level of this comparison it carries the **same free-parameter count as** ΛCDM. (This equal count is at the background/geometry level; the perturbation sector adds one *disclosed* parameter, the braiding fraction b, registered as a pre-committed falsifier measurable at $\sim 5\sigma$ by DESI DR3 + Euclid — not fit to the likelihoods driving this comparison.) At the background level (BAO+SN+BBN) the two models are indistinguishable, $\Delta\chi^2 = -0.43$. With the full likelihoods, the best-fit comparison gives $\Delta\chi^2(\mathrm{SEDE} - \Lambda\mathrm{CDM}) = -3.4$, and converged MCMC chains give a marginalized ΔDIC $= -3.5$ — of which -3.0 is the mean-χ^2 difference and the remaining half-unit an effective-complexity term inside the Monte-Carlo scatter. At equal parameter count SEDE is therefore **statistically consistent with** ΛCDM**, with a mild same-direction pull** (ΔDIC ≈ -3.0, "positive but not strong" evidence) residing entirely in the Planck high-ℓ likelihood, in the direction of Planck's mild lensing-amplitude anomaly. Because that anomaly is itself partly a lite-likelihood systematic, we test the pull directly against the Planck 2018 lensing ($\phi\phi$) likelihood in a full BAO/SN/BBN-anchored joint refit ($\phi\phi$ as a best-fit datum, separate from the DIC chains):

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it erodes the preference from −3.5 to −2.3, consistent with about a third of the gain being the $A_{\rm lens}$ systematic rather than physical lensing — we make no detection claim. This is the SH0ES-free, full-likelihood counterpart of the compressed-CMB preference reported in the companion cosmology paper (there ΔDIC ≈ -3.5 to -4.7, its upper end carried by the contested local-H_0 prior and the compression); the independent full-likelihood test here lands at the conservative end, ΔDIC ≈ -3.0. En route we diagnose and fix a numerical pathology of the stable-parametrization scalar sector (a spurious late-time oscillation seeded at the GR→EFT transition that mimics a large ISW signal), and we validate the corrected pipeline exhaustively. SEDE's discriminating predictions survive intact and are led by the one that does not depend on the solver regime: a **+2–3% scale-independent $f\sigma 8$ enhancement** through the RSD range — fixed by the sub-horizon coupling $\mu_\infty = 1.05$, independent of the tachyonic-window regularization, opposite in sign to what would ease the S_8 tension, and within DESI DR3/Euclid reach — with a corroborating **−7.6% ISW suppression** at $\ell = 2$ (computed below the window and correspondingly more model-dependent). The pipeline is rerunnable from the released configurations and a pinned fork of the Boltzmann code (the chain products and DIC-reduction script are to be committed with the release; see Reproducibility).

Keywords: dark energy theory, cosmological parameters from CMBR, baryon acoustic oscillations, modified gravity

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1. Introduction

SEDE posits that dark energy is the thermodynamic response of the cosmic horizon with a volume-law ("Barrow $\Delta = 1$") entropy, activated by the formation of cosmic structure through a saturation gate $f_{\rm sat}(\sigma^2_{\rm R})$. Its background history is not a fitted w_0–$w_{\rm a}$ curve: given the gate and the entropy law, w(z) is produced by a fixed-point reconstruction (FPAB) whose output — a cubic KGB theory in the stable parametrization ($\Delta M^{*\mathbf{2}}$, $D_{\rm kin}$, $c_{\rm s}^2$), with braiding $\alpha_{\rm B} = b \cdot \Omega_{\rm X}$ and b = 0.20 (the adopted value; the QCD-derived braiding fraction is $b \approx 0.206$) — is then frozen. The model therefore enters a likelihood comparison with **zero extra continuous background parameters** relative to ΛCDM; its dark-energy sector is either right or wrong as a shape.

The companion cosmology paper [1] reports the same mild preference from compressed CMB distance priors combined with a local-H_0 prior (ΔDIC ≈ -3.5 to -4.7, the upper end carried by the contested SH0ES prior and the compression). Compressed priors plus a local-H_0 prior are a weaker foundation than the real likelihoods; the purpose of this paper is to make the test as strong as we can — the same question, asked of the real likelihoods through the real Boltzmann hierarchy, with the full posterior rather than a best-fit point, and SH0ES-free. It returns a result **statistically consistent with** ΛCDM: a marginalized $\Delta\mathrm{DIC} = -3.5$ ($\Delta\langle\chi^2\rangle \approx -3.0$), a mild same-direction pull at the conservative end of the companion's range.

Three results follow. First, the geometry: SEDE $\approx \Lambda$CDM on BAO+SN, as expected for a background that tracks ΛCDM to $|1+\mathrm{w}| \lesssim 0.01$ at low z. Second, the full comparison: SEDE is **statistically indistinguishable from** ΛCDM, with a mild same-direction pull ($\Delta\mathrm{DIC} = -3.5$, $\Delta\langle\chi^2\rangle \approx -3.0$) concentrated in Planck's high-ℓ TTTEEE, where SEDE's enhanced growth supplies extra lensing smoothing in the direction of the well-known A_{lens} tendency [2, 3] — a direction in which that anomaly is itself partly a lite-likelihood systematic, so the origin of the pull is left for the Planck lensing likelihood to decide, not claimed here. Third, the predictions that make the model falsifiable rather than merely fitting, led by the window-independent one: a +2–3% $f\sigma 8$ signal fixed by the sub-horizon coupling at exactly the scales DESI DR3 and Euclid measure, corroborated by $a - 7.6\%$ ISW deficit at the lowest multipoles.

We also report, deliberately and in detail, a numerical failure mode of the public modified-gravity solver in this regime and its resolution (Section 3, Fig. 1): the previously adopted value of the GR-transition redshift produced a grossly corrupted CMB spectrum which briefly led us to retract the ISW prediction; the retraction was wrong, the pathology is understood, and the guarded pipeline cannot silently reproduce it.

2. Model, pipeline and data

Model. FPAB-SEDE [1, 4] at the b = 0.20 fixed point, a cubic kinetic-gravity-braiding theory [5]. The background is supplied to the Boltzmann code as tabulated $\rho_{\mathrm{DE}}(a)$ (`expansion_model = rho_de`) and the perturbation sector as the stable-parametrization [6] Chebyshev coefficients (`gravity_model = stable_params`, $\alpha_{\mathrm{B0}} = 0.139982$; the braiding $\alpha_{\mathrm{B}} = b \cdot \Omega_{\mathrm{X}}$ is the standard density-tracking ansatz, with b = 0.20 in place of the generic 1.5). Both files are frozen inputs, committed with the code fork.

The gate in $\rho_{\mathrm{DE}}(a)$ is built from the *background* ($\mu = 1$) growth amplitude D(z) — a functional of the expansion history, not the solver's fifth-force-enhanced growth — so the tabulated background carries no hidden loop with $\mu(k)$ (the H $\leftrightarrow f_{\mathrm{sat}}(D)$ fixed point is closed separately to 10^{-13}; cosmology paper §2). The effective gravitational coupling at sub-horizon scales is $\mu_\infty = 1.051$ (1.05 to the precision quoted in prose below); the gravitational slip is unity ($c_{\mathrm{T}} = 1$). To forestall a common misreading: SEDE is *holographic* dark energy — $\rho_{\mathrm{DE}}(a)$ enters an otherwise standard-GR Friedmann background as a single added fluid — and mochi-class (a modified-gravity Boltzmann code) is used here only as the perturbation solver for the cubic-KGB completion; SEDE itself is not a modified-gravity theory of the background.

Code. mochi-class [7] (a hi_{class} [8] descendant of CLASS [9]) provides the modified-gravity Boltzmann hierarchy; we drive it through cobaya 3.6.2 [10]. Two pipeline details matter for anyone reproducing this: (i) cobaya must be pointed at the modified-gravity classy build (`path: global`), otherwise it silently loads a vanilla CLASS from its package path and every SEDE evaluation fails; (ii) the perturbation sector requires the GR-transition setting of Section 3. Our fork, including the fix of Section 3 and the frozen model inputs, is pinned at commits `74d4b05` (solver guard) and `b365166` (inputs).

Data. DESI DR2 BAO [11] (`bao.desi_dr2.desi_bao_all`); Pantheon+ SN [12] (`sn.pantheonplus`); Planck 2018 lowl TT and high-ℓ `plik_lite` TTTEEE [13] (clipy implementation, validated against its test point to 4×10^{-9}). Gaussian priors: $\tau = 0.0544 \pm 0.0073$; BBN $\omega_{\mathrm{b}} = 0.02218 \pm 0.00055$ [14] (applied identically to both models). Sampled parameters: $\{H_0,\ \omega_{\mathrm{b}},\ \omega_{\mathrm{cdm}},\ \log A_{\mathrm{s}},\ n_{\mathrm{s}},\ \tau\}$ (six, identical for both models); the $plik_{\mathrm{lite}}$ calibration nuisance A_{planck} is held fixed at 1.0025.

Spectral guard. Every likelihood evaluation passes through an explicit spectral sanity check (a zero-cost cobaya likelihood): first-peak amplitude within a loose physical window, no negative C_ℓ^{TT} at $30 \le \ell \le 2000$, ISW plateau below 5000 μK^2. A pathological sample is rejected at $-\infty$ rather than entering a chain with a merely large χ^2. In the production runs the guard fired on zero samples: the corrected solver never produced a pathological spectrum, so the guard is insurance against regression rather than a filter the results lean on — its silence is the intended outcome, not an unused code path.

3. A solver pathology and its fix

With the previously adopted GR-transition setting `z_gr_smg = 99`, the SEDE lensed D_ℓ^{TT} is ~1500× too large at the first peak and negative at high multipoles (e.g. $D_\ell(800) < 0$) — physically impossible (Fig. 1b), and, if taken at face value, a spurious $\Delta\chi^2$ ~ 10^{13} "exclusion". Perturbation-level diagnosis: at $k \approx 0.02\ Mpc^{-1}$ the metric combination $(\phi + \psi)$ develops a late-time (z < 4) oscillation of amplitude ~19 (physical: 0.52) with ~19 sign changes — a fake ISW source that swamps the low-ℓ spectrum. Two mechanisms compound:

1. **Off-attractor initialization.** At the GR→EFT switch the scalar is initialized at its quasi-static value; when the EFT functions are tiny (early times) the QS formula is ill-conditioned, and at $z_{\mathrm{gr}} = 99$, k = 0.02 it initializes the scalar ~3500× off its attractor ($V_{\mathrm{x}} = 6 \times 10^6$ vs ~1.7×10^3).
2. **A genuine tachyonic window.** The reconstructed scalar sector has QS $mass^2 < 0$ at $k \lesssim 0.05\ Mpc^{-1}$ and $z \gtrsim$ 30–50 (cosmology-dependent); released inside this window, the mode grows regardless of its initial value. (A per-mode deferral of the switch conflicts with the code's global quasi-static scheduling and was abandoned; the pathology is a property of the reconstruction, not of any IC.)

Fix. (i) A guard in the solver: if the QS $mass^2$ at the switch is non-positive, the scalar starts at $\mathrm{x} = \mathrm{x}' = 0$, continuous with the GR-held regime (commit `74d4b05`). (ii) The transition is placed below the tachyonic window: `z_gr_smg = 5`, where $\Omega_{\mathrm{DE}}(z = 5) \approx 1\%$ — no dark-energy physics is sacrificed.

Is the tachyon fatal? A QS $mass^2 < 0$ is a genuine long-wavelength instability — as the $z_{\mathrm{gr}} = 99$ run above shows, a mode released *inside* the window grows without bound and corrupts the spectrum. What we can rule out is that it is a *ghost* or a *gradient* instability, the two short-wavelength catastrophes that would condemn the EFT outright: across the entire frozen reconstruction ($z \approx 150 \to 0$) the stable-parametrization inputs have a strictly positive no-ghost kinetic term, $D_{\mathrm{kin}} > 0$ everywhere ($10^{-6} \to 1.4$), a strictly positive squared sound speed, $c_{\mathrm{s}}^2 > 0$ everywhere (0.018–0.18), and a non-running Planck mass, $\Delta M^2 = 0$. *The instability is therefore tachyonic — a Jeans-type long-wavelength growth that these no-ghost/no-gradient conditions do* not* by themselves bound — rather than a short-wavelength blow-up. We

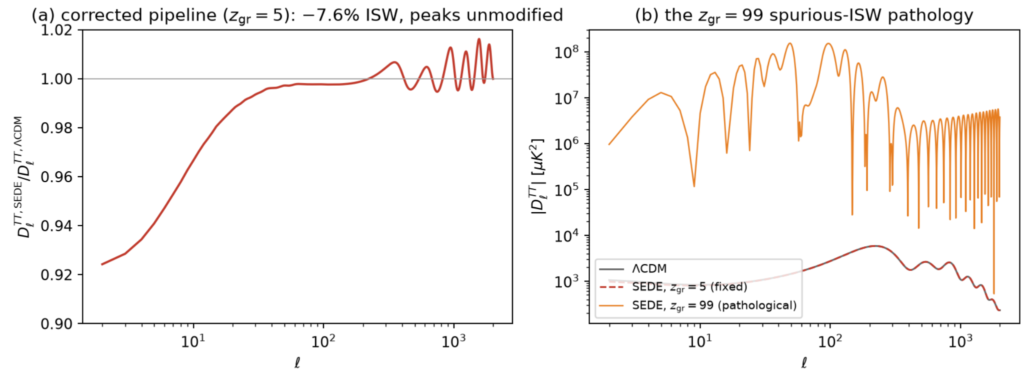

Figure 1: (a) The corrected SEDE/ΛCDM temperature-spectrum ratio ($z_{\mathrm{gr}} = 5$): the -7.6% ISW deficit at $\ell = 2$, recovering to within 0.5% of unity by $\ell \approx 50$, with the acoustic peaks unmodified. (b) The same model computed at $z_{\mathrm{gr}} = 99$: the spurious scalar oscillation injects a fake ISW that corrupts the spectrum by up to three orders of magnitude.

do not claim it is harmless; we regularize it by placing the GR→EFT transition *below* the window (where μ_∞ and the ISW are computed) and seeding the mode at zero. Whether the window is a real feature of the reconstructed EFT or an artifact of the Chebyshev inversion of the stable parametrization — and hence whether the below-window observables inherit any residual contamination — remains open (§7).

Validation. The corrected spectrum is converged and setting-independent: $D_\ell(220)$ and $C_\ell(2)$ agree to $\leq 3 \times 10^{-5}$ across $z_{\mathrm{gr}} \in 3, 5, 10, 20$; a 49-point grid over $H_0 \in [62,\, 75] \times \omega_{\mathrm{cdm}} \in [0.104,\, 0.140]$ ($\Omega_{\mathrm{m}} \approx 0.27$–$0.42$) is sane at every point, as are four hostile corners with $A_{\mathrm{s}}/n_{\mathrm{s}}$ excursions. The primary acoustic peaks are unmodified (ratio 0.9999 to ΛCDM at $\ell = 220$), as they must be — recombination is in the GR regime (Fig. 1a). The ISW suppression — $C_\ell^{\mathrm{TT}}(2)$ ratio $= 0.924$ from the released finite-k pipeline (`sede_fsigma8_isw_finitek.py`), 0.9236 from the z_{gr}-independent stage-2 cross-check, *i.e.* -7.6% either way — agrees with the model's original finite-k computation; the prediction is real, the $z_{\mathrm{gr}} = 99$ spectra were not.

4. Background-level comparison: BAO + SN (+ BBN)

Direct multi-start minimization (Nelder–Mead $\to$ Powell; four starts agreeing to better than 10^{-3}) over $\{H_0, \omega_b, \omega_{cdm}\}$:

model	χ^2	χ^2_{BAO}	χ^2_{SN}	H_0	ω_{cdm}	Ω_m
ΛCDM	1416.18	10.89	1405.29	68.76	0.1215	0.305
SEDE	1415.75	10.75	1405.00	69.13	0.1248	0.306

$\Delta\chi^2 = -0.43$ (this is the BAO+SN+BBN background statistic; the companion's *full-CMB profiled* fit lands on the same number coincidentally — a different test): geometrically indistinguishable, with SEDE preferring H_0 higher by 0.4. This is the expected null — SEDE's discriminating physics is in the perturbations, not the distances.

5. Full-likelihood comparison

Best fit. Multi-start scipy minimization over the five non-prior-pinned parameters (the cobaya Nelder-Mead wrapper stalls at its reference simplex in this setup; the direct minimization is robust, both starts per model agreeing to < 0.1):

model	χ^2 total	BAO	SN	lowl TT	TTTEEE-lite	H_0	ω_{cdm}	n_s
ΛCDM	2023.61	11.53	1406.00	22.86	583.22	68.53	0.1185	0.9682
SEDE	2020.18	11.19	1406.70	22.38	**579.91**	68.86	0.1193	0.9664

$\Delta\chi^2(\text{SEDE} - \Lambda\text{CDM}) = -3.4$, dominated by the high-$\ell$ TTTEEE term (-3.3). (This minimization predates the BBN prior of Section 2 and fixes τ and A_{planck}: ω_b sits at its box edge, 0.0225, for *both* models — identical treatment, so the *model difference* is a fair comparison, but this best fit lives on a different likelihood than the with-BBN, τ-sampled MCMC below, so the two should be read as separate checks of the same direction, not as the same statistic evaluated three ways.) We emphasize an instructive false trail: evaluated at $\Lambda\text{CDM}'s$ own best fit, SEDE's high-ℓ χ^2 is ˜200 *worse* — the imprint of $\mu_\infty = 1.05$ over-lensing the damping tail. That apparent tension is an artifact of the comparison point: the joint refit absorbs it entirely through small shifts ($n_s - 0.002$, $\omega_{cdm} + 0.0008$, $H_0 + 0.33$), leaving a net *gain*. We flag the fragility honestly: $a - 3.4$ that is the residual of a ~200-unit cancellation along the $\omega_{cdm}/H_0/n_s$ degeneracy is sensitive to the joint minimum for *both* models, and our Nelder–Mead tolerance (both starts agreeing to < 0.1) is not far below the signal; the marginalized ΔDIC (below) is the robust statement, the best-fit -3.4 the fragile one. Physically, SEDE's extra lensing points the *same way as* Planck's mild $A_{lens} > 1$ tendency [2, 3] — which is precisely the concern: $A_{lens} > 1$ is at least partly a lite-likelihood systematic — the peak-smoothing excess is not seen in the direct lensing reconstruction [15, 16, 17] and it relaxes toward unity in PR4/CamSpec and with ACT DR6 [18] — so a preference concentrated in exactly that channel could be *absorbing* that systematic rather than detecting lensing physics. Distinguishing the two requires the Planck lensing ($\phi\phi$) likelihood, which the lensing-shaped gain specifically predicts (§7) — until it is run, the physical interpretation of the -3.5 is provisional. A_s alone cannot mimic the effect (lowering A_s degrades the primary peaks).

Marginalized (production MCMC). Converged chains for both models ($R-1$ means: 0.014 / 0.015; bounds < 0.11; 13.0k / 19.6k accepted samples; learned proposal seeded from the Planck `base_plikHM_TTTEEE_lowl_post_BAO` covariance; spectral guard active throughout, zero rejections):

	ΛCDM	SEDE
H_0	68.54 ± 0.29	68.89 ± 0.30
ω_{cdm}	0.11847 ± 0.00063	0.11922 ± 0.00066
n_s	0.9685 ± 0.0032	0.9665 ± 0.0033
$10^9 A_s$ (from log A)	2.113 ± 0.031	2.112 ± 0.032
$\langle\chi^2\rangle$	2028.1	2025.1
p_D	6.8	6.3
DIC	2034.9	**2031.4**

$$\Delta\text{DIC}(\text{SEDE} - \Lambda\text{CDM}) = -3.5 \tag{1}$$

With identical priors and the same parameter count, this is a marginalized, real-likelihood statement that SEDE fits the joint dataset mildly better — "positive but not strong" evidence in the DIC 2–5 band [19] (we attach no σ-level: the models are non-nested and ΔDIC is not χ^2-distributed). The p_D difference (6.8 vs 6.3) is within the Monte-Carlo scatter of p_D at these chain lengths and the posterior widths are nearly identical, so we read this *pre-lensing* number as $\Delta\mathrm{DIC} \approx \Delta\langle\chi^2\rangle \approx -3.0$ rather than resting on a half-unit of effective-complexity difference. The best-fit -3.4 and the marginalized -3.5 point the same way but are computed on different likelihoods (above); we treat the marginalized ΔDIC as the pre-lensing headline — folding in the direct $\phi\phi$ datum below reduces it to -2.3, which is the number we ultimately quote. (The companion's compressed CAMB-in-the-loop pipeline gives a related full-CMB $\Delta\chi^2 = -3.17$ on its wider DESI + SN + CC + $f\sigma 8$ data vector; the -3.4 here is the independent mochi-class best-fit on the DESI+SN+Planck vector.) SEDE's posterior pulls H_0 upward by $+0.35$ (0.8σ of the combined width) — the right direction for the H_0 tension, at an inconsequential magnitude. The joint posterior shifts are shown in Fig. 2.

The Planck-lensing ($\phi\phi$) cross-check. The preference above lives entirely in the high-ℓ TTTEEE smoothing, in the direction of Planck's A_{lens} tendency — which is partly a lite-likelihood systematic (above, and §7). The direct Planck 2018 lensing ($\phi\phi$) likelihood [20], whose reconstructed amplitude sits at ≈ 1.0 rather than the TT-derived ≈ 1.2, decides whether SEDE's extra smoothing is physical lensing or an absorbed systematic — the same peak-smoothing-vs-reconstruction consistency test used model-independently by Planck and by Motloch & Hu [15, 16], here applied to SEDE — so we ran it. We refit *both* models over $\{H_0, \omega_{\mathrm{b}}, \omega_{\mathrm{cdm}}, A_{\mathrm{s}}, n_{\mathrm{s}}\}$ to the full DESI DR2 BAO + Pantheon+ + BBN + lowl TT + TTTEEE data — once without and once with `planck_2018_lensing.native` (CMB-marginalized). Without $\phi\phi$ the joint preference is $\Delta\chi^2 = -3.5$ (reproducing the best fit and the DIC); **with** $\phi\phi$ **it is** $\Delta\chi^2 = -2.3$. SEDE fits $\phi\phi$ *worse* than ΛCDM by $+1.2$ ($\chi^2_{\phi}\phi = 10.0$ vs 8.8), because $\mu_\infty = 1.05$ boosts $C_\ell^{\phi\phi}$ above the measured amplitude; the direct lensing datum thus **erodes the preference from** -3.5 **to** -2.3 ($\approx\frac{1}{3}$; 66% survives), so about a third of the high-ℓ TTTEEE gain is *consistent with* the absorbed A_{lens} systematic rather than physical lensing (the test cannot separate an absorbed systematic from μ_∞ over-predicting real lensing). Because BAO + SN + BBN pin the background density ($\omega_{\mathrm{cdm}} \approx 0.119$ in both fits), this anchored joint refit is the honest number: a CMB-primary-only refit that frees ω_{cdm} gives a smaller penalty ($+0.3$) *only* by letting $\Lambda\mathrm{CDM}'s$ ω_{cdm} drift to an unphysical 0.120, and an A_{s}-profiled evaluation at the primary optimum gives -1.9 as a conservative (off-minimum) upper bound — the anchored joint refit at -2.3 supersedes both. The $\phi\phi$ likelihood is CMB-marginalized, so it shares some Planck TT information and is not a fully

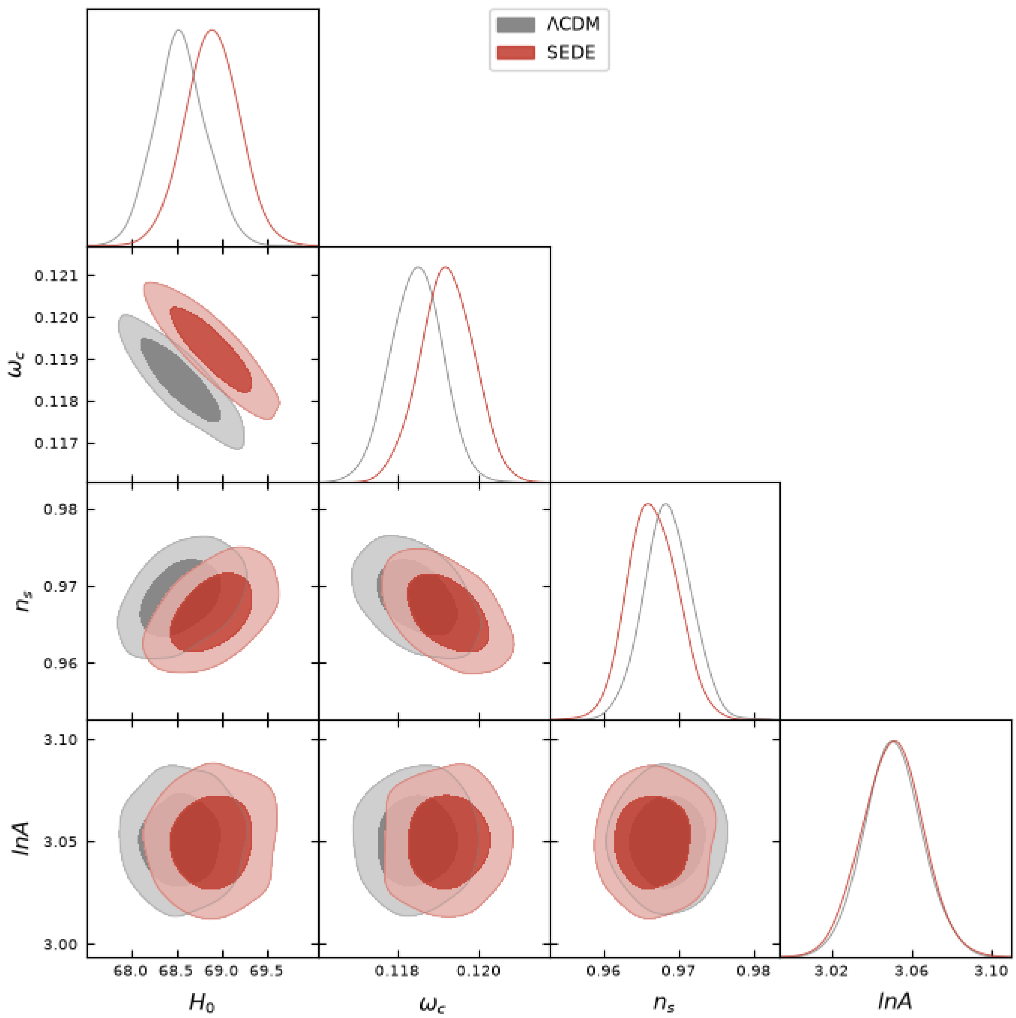

Figure 2: Marginalized posteriors (68% and 95%) for H_0, $\omega_{\rm cdm}$, $n_{\rm s}$ and $\log A_{\rm s}$: ΛCDM (grey) vs SEDE (red). The coherent small shifts (H_0 up, $n_{\rm s}$ down, $\omega_{\rm cdm}$ up) are how SEDE accommodates its extra lensing.

independent datum, and $\chi^2_\phi\phi(\Lambda\text{CDM}) \approx 8.8$ is consistent with (not a proof of) a well-calibrated ∼9-bandpower likelihood. This is the direct evidence behind the "consistent with ΛCDM, mild pull" reading (repro `phiphi_joint_refit.py`; earlier estimates and caveats in `results/phiphi_test_result.md`).

6. Predictions

The comparison above is a consistency statement; the model's falsifiable content is forward-looking. We separate it by robustness: the growth signal (1) is fixed by the sub-horizon coupling μ_∞, set above the tachyonic window and independent of the solver's GR-transition regularization — this is the **primary pre-registered test**; the ISW signal (2) is computed *below* the window and is correspondingly more model-dependent, a **corroborating** rather than a primary falsifier.

1. **Growth (primary falsifier).** $P_{\text{SEDE}}/P_\Lambda\text{CDM} = +2.6\%$, flat across k = 0.01–1 h/Mpc; $\sigma 8$ ratio +1.3%; and a **+2–3% scale-independent $f\sigma 8$ enhancement through the RSD range** (Fig. 3; from the sub-horizon coupling $\mu_\infty = 1.05$, fixed by α_{B0} and independent of the tachyonic-window regularization; k-table: +2.9/+2.3/+2.0/+1.7% at z = 0/0.3/0.5/1 for k = 0.1 h/Mpc). Current $f\sigma 8$ errors are 5–8%; DESI DR3 and Euclid reach the 2–3% level. **The sign is opposite to what would ease the S_8 tension**, and opposite to the mild *suppression* of growth ($\gamma > 0.55$) current data prefer [21] — this prediction is a genuine risk, not an accommodation. An $f\sigma 8$ measured *not* enhanced at this level falsifies the $\mu(k)$ sector (pre-registered as P5 — the growth+ISW signature — of the program's frozen P1–P20 falsifier matrix; not to be confused with the cosmology paper's *internal* prediction numbering, whose P3–P5 are the withdrawn SOC layer).
2. **ISW (corroborating).** C_ℓ^{TT} ratio 0.924 at $\ell = 2$, 0.987 at $\ell = 20$, within 0.5% of unity by $\ell \approx 50$: $a - 7.6\%$ low-ℓ deficit, cosmic-variance-limited but contributing to low-ℓ TT likelihoods and to ISW–LSS cross-correlations. Because it is computed below the tachyonic window (§3), this prediction is more model-dependent than the growth signal and should be read as corroborating it, not as an independent primary falsifier. This is also where FPAB-SEDE separates cleanly from the cubic-Galileon models excluded at ∼7–8σ by the ISW–galaxy

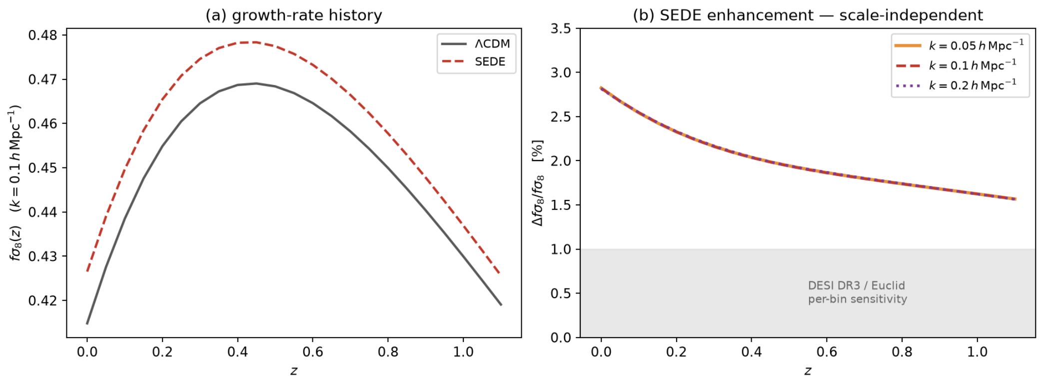

Figure 3: The growth prediction. (a) $f\sigma8(z)$ at k = 0.1 h/Mpc for ΛCDM and SEDE at the shared best-fit cosmology. (b) The fractional SEDE enhancement at k = 0.05, 0.1 and 0.2 h/Mpc — the three curves lie on top of one another, which is the point: the enhancement is scale-independent through the RSD range, entering the DESI DR3/Euclid sensitivity band. Underlying data: `src/fsigma8_prediction.json`, regenerated by `src/sims/sede_fsigma8_isw_finitek.py`.

cross-correlation [22]: that exclusion targets *self-accelerating* Galileons, whose kinetic term drives the acceleration and flips the sign of the ISW × *galaxy* cross-power (anti-correlation). FPAB-SEDE is **not** self-accelerating — the acceleration is carried by the background $\rho_{\mathrm{DE}} \propto H f_{\mathrm{sat}}$, and the braiding is a weak ($b \approx 0.2$, $\mu_\infty = 1.05$) sub-horizon correction — so its ISW × *galaxy* cross-power keeps the ΛCDM sign and is mildly *enhanced* (+13% at $\ell = 2$, with the −7.6% temperature-auto deficit above being the standard auto-vs-cross sign structure of a stronger late-ISW source; cosmology paper §7). SEDE therefore evades the Renk et al. exclusion by construction rather than by tuning.

3. **DE clustering.** $\delta_{\mathrm{DE}}/\delta_{\mathrm{m}} \lesssim 7 \times 10^{-5}$ at k = 0.1 h/Mpc (horizon-confined): any detected gate–matter cross-correlation at RSD scales kills the realization outright.

7. Limitations

- ΔDIC $= -3.5$ (robust part $\Delta\langle\chi^2\rangle \approx -3.0$) is positive, not decisive; the data are statistically consistent with ΛCDM and we make no detection claim.

- **We ran the decisive test — the Planck lensing** ($\phi\phi$) **likelihood — and it pulls the preference down.** The entire preference sits in the high-ℓ TTTEEE term, where Planck's $A_{\rm lens} > 1$ tendency lives. Adding the direct $\phi\phi$ likelihood (`planck_2018_lensing.native`, reconstructed amplitude ≈ 1.0) in a full BAO/SN/BBN-anchored joint refit erodes the preference from -3.5 to $\mathbf{-2.3}$ (§5): about a third of the TTTEEE gain is consistent with the absorbed $A_{\rm lens}$ systematic rather than physical lensing. Repeating against a PR4/CamSpec or ACT high-ℓ likelihood (where $A_{\rm lens} \approx 1$), and the full multi-nuisance plik (vs the lite used here) with $\phi\phi$ in the marginalized chains, are the remaining checks — expected to point the same way. $The - 3.5$ is thus a pre-$\phi\phi$ upper bound; the direct lensing datum reads the preference down to -2.3.
- No w0waCDM baseline is run here: SEDE's selling point (same physics, zero extra parameters) is best shown against the 2-extra-parameter parametric alternative; the (w_0,w_a) placement is in the companion ($\sim$2.7σ from the DESI point, $\sim$0.5σ from ΛCDM), and a direct w0waCDM ΔDIC is deferred.
- One Boltzmann implementation (one fork) computed everything; an independent EFT code (e.g. EFTCAMB [23]) reproducing the μ_∞/ISW numbers would remove a single-implementation systematic.
- The solver operates *below* the reconstruction's tachyonic window rather than through it. The window is a genuine long-wavelength (Jeans-type) instability — though *not* a ghost or gradient one ($D_{\rm kin} > 0$ and $c_{\rm s}^2 > 0$ throughout, §3), which is why regularizing it by seeding below the window is legitimate rather than a suppression of healthy physics; whether the window itself is a real feature of the reconstructed EFT or an artifact of the Chebyshev inversion deserves a dedicated study. The sub-horizon growth signal (μ_∞, $f\sigma 8$) is set *above* the window and is insensitive to this question; the ISW and the lensing-shaped -3.3, computed *below* it, are conditional on it — one further reason we take the $f\sigma 8$ prediction, not the ISW or the -3.5, as the primary falsifiable claim.
- The background and stability inputs are frozen at $\mathrm{b} = 0.20$; the braiding fraction is a free parameter of the wider program (forecast measurable at ~5σ by DR3+Euclid), not varied here.
- **Background-only H(z) cannot arbitrate, and this is structural rather than a data limitation.** Against the model-independent 31-

point cosmic-chronometer compilation SEDE fits with $\Delta\chi^2 = +0.04$ relative to flat ΛCDM at equal parameter count, and the model's background shape is degenerate with an (Ω_m, H_0)-shifted ΛCDM at the ~0.2% level over $z \in [0, 2]$ (an oΛCDM fit to the SEDE curve returns Ω_k,eff $= 0.000$; script in the cosmology companion, `run_deficit_data.py`). No realistic expansion-history precision changes this — which is a further reason the discriminating claims of this paper are placed in the growth ($f\sigma 8$), tensor (Ξ_0), and Δ-exponent sectors, where the gate's imprint is physical rather than reparameterizable.

8. Conclusions

At equal parameter count against the real likelihoods, structure-gated dark energy is **statistically consistent with** ΛCDM, with a mild same-direction pull (marginalized $\Delta\mathrm{DIC} = -3.5$, of which the robust part is $\Delta\langle\chi^2\rangle \approx -3.0$ — "positive but not strong" evidence). The pull sits entirely in the lensing-scale smoothing of the Planck damping tail, in the direction of Planck's A_lens tendency; because that tendency is itself partly a lite-likelihood systematic, we ran the decisive test — the direct Planck lensing ($\phi\phi$) likelihood (§5) — in a full BAO/SN/BBN-anchored joint refit it pulls the preference down from −3.5 to **−2.3**: about a third of the gain is consistent with the absorbed A_lens systematic rather than physical lensing. We make no detection claim. What makes the model falsifiable on a short horizon is not this pull but its forward predictions, led by the one independent of the solver regime: a +2–3% sub-horizon $f\sigma 8$ enhancement that DESI DR3/Euclid can confirm or kill, corroborated by an ISW deficit, and a strict null in gate–matter correlations at sub-horizon scales. Within these caveats the compressed-CMB comparison of the companion paper is reproduced — SH0ES-free — by this full-likelihood analysis: both give a mild, same-sign, non-decisive pull, the full-likelihood $\Delta\mathrm{DIC} \approx -3.0$ sitting at the conservative end of the companion's $\Delta\mathrm{DIC} \approx -3.5$ to -4.7 range (whose upper end is carried by the contested SH0ES prior — ~40% of the companion's evidence — and the compression-vs-full dilution the companion itself reports, $-4.68 \to -3.17$). Having now folded in the direct Planck lensing datum, we regard −2.3 as the honest post-$\phi\phi$ preference — mild, same-sign, and firmly non-decisive.

Reproducibility

The configurations, minimizers, the spectral guard, the diagnostic scripts and the chains' SHA-256 checksums are in the `sede-observational-tests` repository (a tagged release is archived at Zenodo, DOI 10.5281/zenodo.21525527); the production chain files, the minima, and the chain→($\langle\chi^2\rangle$, $p_{\rm D}$, DIC) reduction script are to be committed with the release, so that the headline ΔDIC is auditable and not only rerunnable. The Boltzmann engine is the pinned mochi-class fork (`74d4b05` solver guard; `b365166` frozen FPAB inputs; build: `rm python/classy.cpp && pip install . --no-build-isolation`). The fork location is set via `MOCHI_DIR`, the frozen SEDE inputs are vendored in `src/mcmc/inputs/`, and likelihood data install via `cobaya-install` from the included `likes_install.yaml` (full setup in `src/README.md`). The growth/ISW predictions ($\sigma 8$, $f\sigma 8$, the $C_\ell^{\rm TT}(2) = 0.924$ ISW ratio) regenerate directly from `src/sims/sede_fsigma8_isw_finitek.py`; the production chains were run with the exact `lcdm_mc.yaml` / `sede_mc.yaml` released here (Rminus1 stop 0.02; the loose $Rminus1_{\rm cl,stop}$ 0.2 on the tails should be tightened for the released run). The Planck-lensing ($\phi\phi$) test of §5: the definitive BAO/SN/BBN-anchored joint refit is `src/mcmc/phiphi_joint_refit.py` (needs the full DESI DR2 BAO + Pantheon+ + Planck primary + lensing data via `cobaya-install`), giving $\Delta\chi^2 = -3.5 \to -2.3$ with $\phi\phi$; the earlier $A_{\rm s}$-profiled and CMB-only estimates (`phiphi_As.py`, `phiphi_fullrefit.py`) and all caveats are in `src/results/phiphi_test_result.md`.

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